Per reduïr les energies (energia total = suma d'energies):

$$U^*_{ArAr} = 4\frac{\epsilon_{Ar}}{\epsilon_{ArKr}}\left[\left(\frac{\sigma_{Ar}}{\sigma_{ArKr}}\frac{1}{r^*}\right)^{12} - \left(\frac{\sigma_{Ar}}{\sigma_{ArKr}}\frac{1}{r^*}\right)^{6}\right] \qquad U^*_{ArKr} = 4\left[\left(\frac{1}{r^*}\right)^{12} - \left(\frac{1}{r^*}\right)^{6}\right]$$

(id. pel KrKr)

U^{tail} tindrà 3 contribucions (A,B = Ar, Kr), calculades amb:

$$U^{tail}_{AB} = 2\pi N_A \rho_B \int_{r_c}^{\infty} \sum_{A,B} N_A N_B U_{AB}(r)\; r^2 \,\mathrm{d}r$$

P^{tail} tindrà les 3 contribucions calculades de forma anàloga amb:

$$P^{tail}_{AB} = -\frac{2\pi}{3V} \int_{r_c}^{\infty} \sum_{A,B} N_A N_B \frac{dU_{AB}(r)}{dr}\; r^3\, dr$$